

Maximizing EMEA Profits from 5.46 % to 20.61%

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Word Count : 1429 words

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Executive summary

In the dynamic economic condition, profit margin becomes key to measure business performance, primarily when higher sales doesn't necessarily translate into high profit. Looking at the global performance, EMEA is struggling on its profit margin while the other markets are performing well. As the EMEA manager, further analysis of the issue must be conducted, along with solutions to tackle it.

This report highlighted the main challenges in EMEA that stated below.

- EMEA has the lowest profit margin among all markets that are driven by Turkey.
- Turkey has the largest customer base in EMEA that leads to second highest sales. However, it generates negative profit.
- Turkey has an inverse relationship between sales and profit.
- Turkey has a 60% discount across all subcategories, while other EMEA countries have a 4.33% discount on average.
- Discount in Turkey is not explaining the sales, instead monthly seasonality does.
- Turkey's average shipping cost per unit is high in tables, appliances, and phones sub-category.
- First class mode accounts for the highest shipping cost and shipping-to-sales ratio per unit.
- Shipping cost also varies by state.

Through these findings, reducing discount to the average is recommended. Lowering the shipping cost for tables, appliances, and phones sub-category is necessary. Especially if critical or high order priority and the shipping happens in states with high shipping costs.

By implementing the recommendations, EMEA profit margin is expected to increase from 5.46% to 20.61%. EMEA will become the highest profit margin across the market.

Introduction

The purpose of analysis is to gain beneficial insight for business and discover problems to improve overall performance. This report focuses on EMEA low profit margin with in-depth analysis of Turkey. The objective is to maximize EMEA potential profit by resolving the root cause and improve its profit margin by 15.15%. Actionable recommendations will be provided at the end of analysis that transform EMEA position from the least profitable into the most profitable across markets.

The superstore sample dataset is used in this report which contains order information, from sales, quantity, shipping cost, to profit. These data are considered as numerical data. Furthermore, it provides geographical data such as region, market, country, and states that allows analyzing by location. The order id, customer id, customer name, are the key identifiers. As well as category, sub-category, and product name which are categorical data. There's also ordinal data like order priority.

1. 0 Market Performance Overview

Market Overview	Turkey Details	Discount Analysis	Shipping Cost Analysis	Projected result
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Market Performance Overview

EMEA low profit margin driven by Turkey

Sales & Profit by Market

Profit is indicated by color, green (high), red (low).

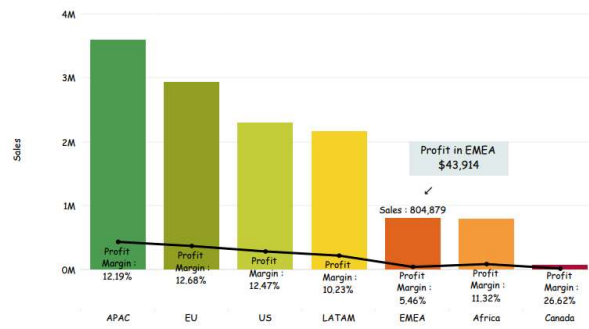


Figure 1.1 Sales & Profit by Market

Top 5 Countries in EMEA by Sales

Profit Margin displayed as percentage (%)

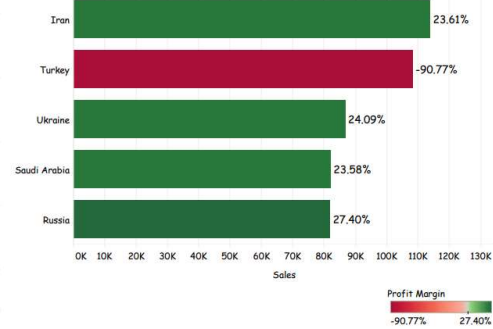


Figure 1.2 Top Five Countries in EMEA by Sales

Number of Customers & Profit by Country in EMEA

Size represents distinct customer count

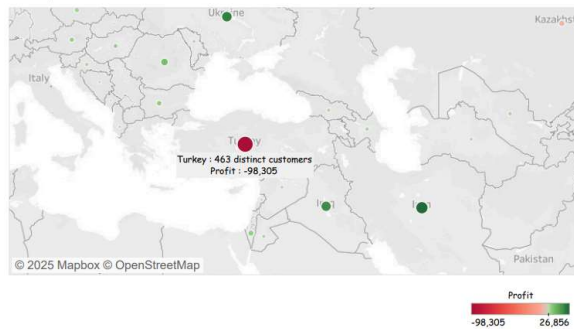


Figure 1.3 Number of Customers & Profit by Country in EMEA

Profit Analysis in EMEA by Excluding Turkey

without Turkey, EMEA profit margin increased from 5.46% to 20.42%

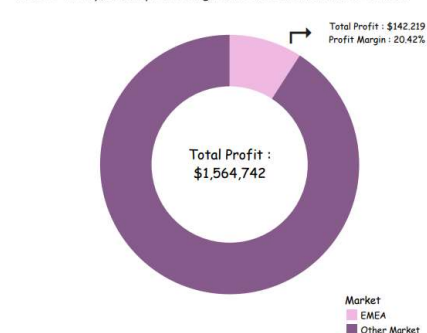


Figure 1.4 Profit Analysis in EMEA by Excluding Turkey

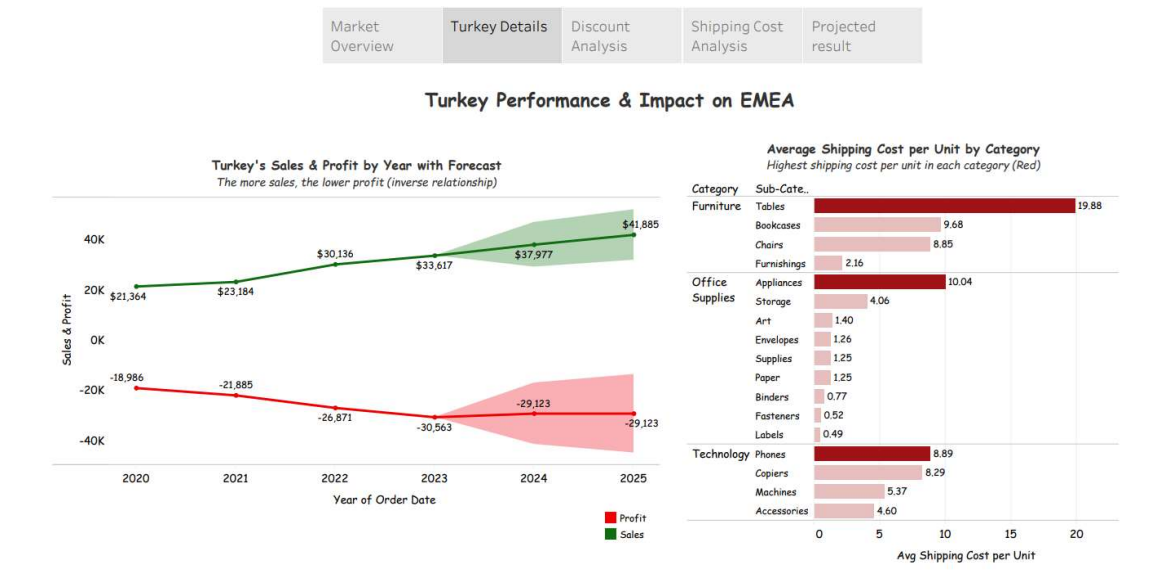
Out of seven markets, EMEA has the lowest profit margin around 5.46% with \$43,914 in profit. This happened mainly due to Turkey's weak performance. Figure 1.2 shows the top five countries with highest sales in EMEA. Although Turkey has the second-highest sales, it generates a -90.77% profit margin.

Figure 1.3 displays Turkey as the largest customer base with 463 distinct customers, which aligns with high sales. However, Turkey still suffers \$98,305 losses, which is the greatest loss in EMEA.

To strengthen the argument that Turkey has dragged down EMEA overall performance, figure 1.4 illustrates how the profit margin when Turkey is excluded from the analysis.

Without Turkey, EMEA profit margin has increased to 20.42%. It's improved significantly by 14.96%. Profit rose to \$142,419.

2.0 Turkey Performance & Impact on EMEA



To get detailed insights of Turkey performance, figure 2.1 illustrates the sales and profit from 2020 to 2023 with its forecast in the next two years. Turkey's sales have grown from \$21,364 to \$33,617, but profit continues to generate bigger losses. This shows the inverse relationship, which is an uncommon scenario. Without taking any action, in 2024 and 2025, it will lead to more losses.

Figure 2.2 shows the average discount and profit margin in EMEA by comparing with and without Turkey analysis. With Turkey, the highest profit margin in EMEA is fasteners subcategory, 14.33%. The average discount is around 19.29% across sub-category. By excluding Turkey from analysis, the lowest profit margin is tables, 17.70%. While the average discount is only 4.33%. This indicates Turkey has a higher discount that lowered down the overall EMEA profit margin.

Figure 2.3 explains the average shipping cost across sub-category in Turkey. Tables, appliances, and phones incur the largest shipping cost in the furniture, office supplies, technology category respectively. Tables and appliances take at least twice the amount of the other subcategory within the same category.

3.0 Discount Analysis in Turkey

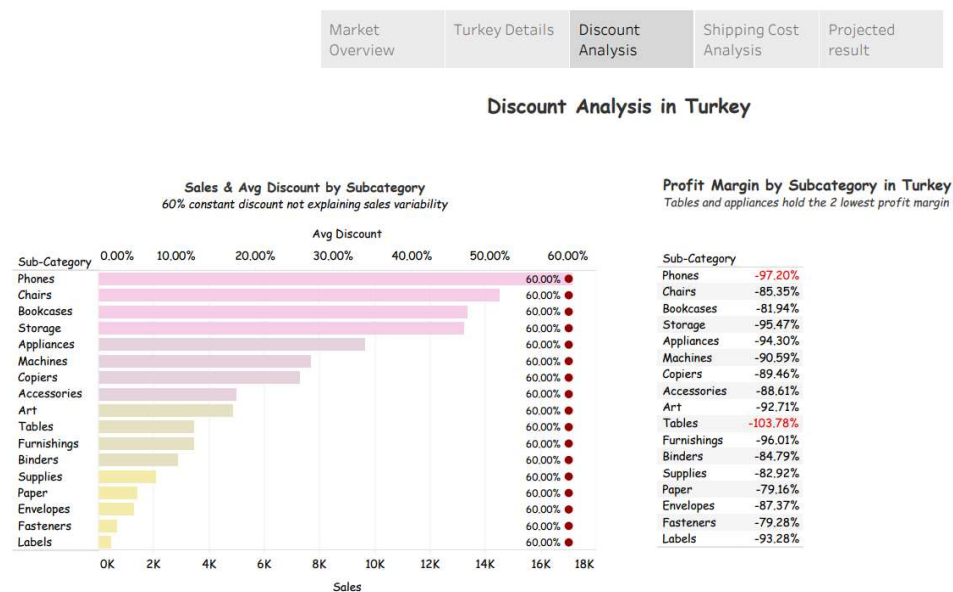


Figure 3.1 Sales & Average Discount by Sub-category

Figure 3.2 Profit Margin by Sub-category

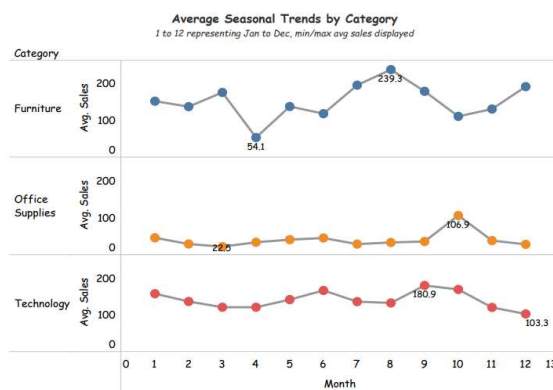


Figure 3.3 Average Monthly Seasonal Trends by Category

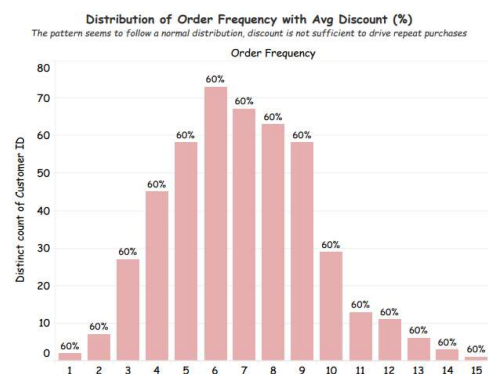


Figure 3.4 Distribution of Order Frequency with Average Discount

Due to Turkey's significant impact on EMEA overall discount, figure 3.1 displays how the average discount in Turkey. It shows that discount is constant at 60% in all sub-category, while the sales vary. This indicates discount isn't explaining variation in sales. In addition, Turkey's discount is at least tenfold of other EMEA countries' discount, highlighting how significantly high it is. When the discount is too high, it's no longer an external cue for customers to purchase.

Figure 3.2 illustrates the profit margins in each subcategory, where all of them experiencing negative profit margin. The main cause is due to the high discount which is explained in figure 3.1. Beside that, tables and phones are the two lowest profit margins, -103.78% and -97.20% respectively. These subcategories are the highest shipping cost in the respective category (figure 2.3). This highlighted there's potential correlation between shipping cost and profit margin which require further analysis.

Figure 3.3 displays the monthly seasonal trends in each category that explains why the sales vary. Office supplies category hit the highest average sales in October while technology reached its peak in September. These correspond with the back-to-school period during September to October. Furniture category reaches its peak in August, while the lowest is in April. Office supplies had lowest sales in March.

Figure 3.4 shows the distribution of order frequency which looks like normal distribution. Discount is constant at 60% while customer retention is vary. Discounts unlikely to encourage customers to purchase again.

4.0 Shipping Cost Analysis in Turkey

Market Overview	Turkey Details	Discount Analysis	Shipping Cost Analysis	Projected result
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Shipping Cost Analysis in Turkey

In tables, appliances, and phones subcategories

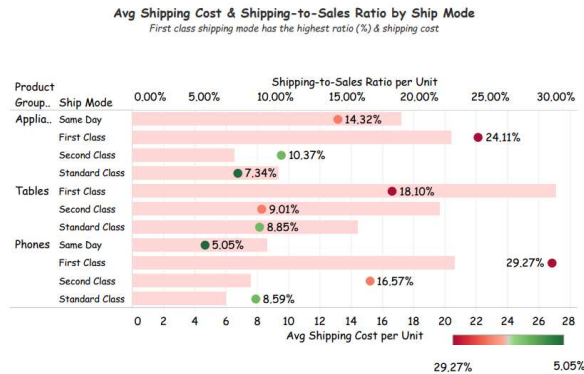


Figure 4.1 Average Shipping Cost & Ship-to-sales ratio by Ship Mode

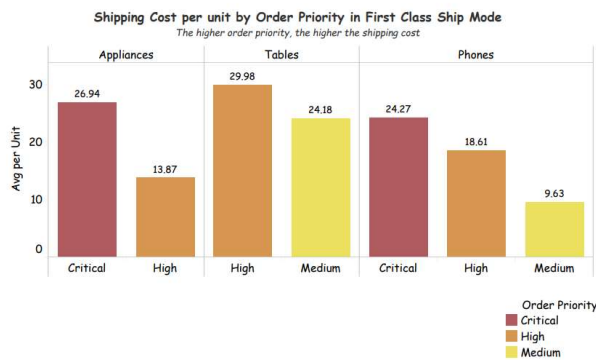


Figure 4.2 Shipping Cost per Unit by Order Priority

Shipping Cost Vary by State

Even with the same product, year, order priority, ship mode, shipping costs differ

Product Name	Year of S.	Order Priority	Ship Mode	State	Quantity	Sales	Shipping Cost
Apple Speaker Phone, Cordless	2022	Medium	Standard Class	Antalya	2.0	100.9	9.6
				Bursa	2.0	100.9	11.6
Samsung Signal Booster, with Caller ID	2022	Medium	Standard Class	Bursa	4.0	218.5	1.7
				Izmir	1.0	54.6	4.7

Figure 4.3 Shipping Cost Vary by States

Average Shipping Cost by State in Appliances, Tables, Phones

Izmir ranks in the top 3 highest avg shipping cost in Tables & Phones subcategories

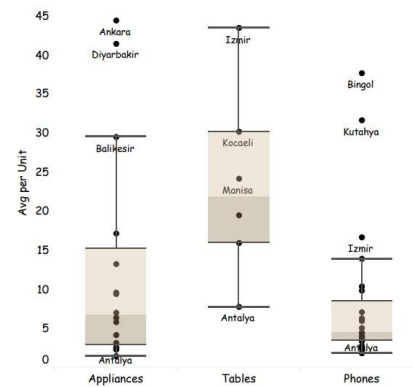


Figure 4.4 Average Shipping Cost by States

Figure 4.1 illustrates the average shipping cost per unit in tables, phones, and appliances sub-category that varies by ship mode. It shows that first class mode poses the highest cost. However, high costs are not necessarily an issue if it's accompanied by higher sales. Therefore, the observation is validated further by the high ship-to-sales ratio per unit in first class mode, which means there's a bigger portion of sales revenue being used for shipping cost rather than translating into profit.

Detailed analysis in first class mode based on order priority is shown by figure 4.2. The higher the priority, the larger the shipping cost is.

Shipping cost has also varied by state. Figure 4.3 displays “Apple Speaker Phone, Cordless” that have been sold in Antalya and Bursa, with the same quantity, order priority, year, quantity, and shipping mode. However, the shipping cost between these states are differ. As

well as “Samsung Signal Booster, with Caller ID”, where the quantity purchased is four times more than in Izmir. However, shipping cost is higher for Izmir, which strongly highlighted how Izmir has a higher shipping cost than Bursa. While Bursa shipping cost is higher than Antalya.

Figure 4.4 illustrates the states with higher shipping cost. Izmir spent the highest shipping cost in appliances, and the third-highest in phones. Antalya consistently has low shipping costs. This reinforces figure 4.3 where Izmir has higher shipping cost than Antalya. Other states such as Ankara, Diyarbakir, Balikesir, Kocaeli, Manisa, Bingol, Kutahya are considered as high shipping cost states.

5.0 Projected Result

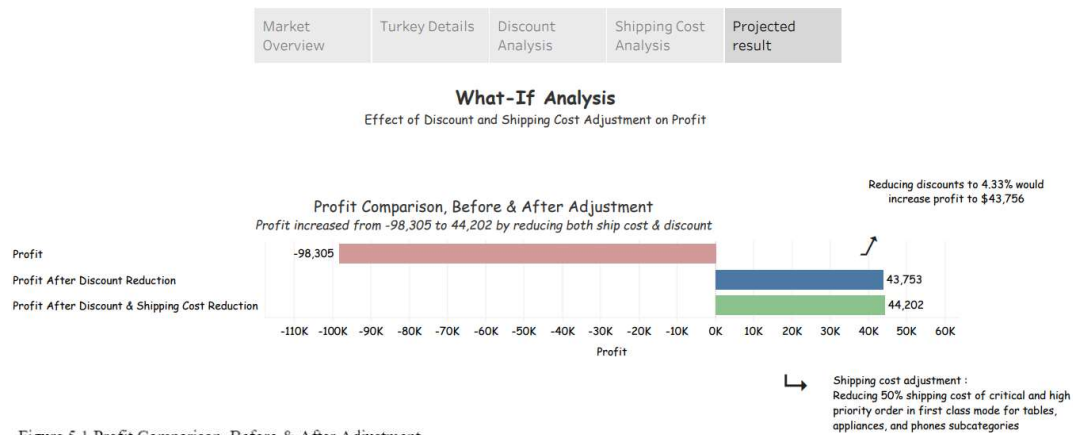


Figure 5.1 Profit Comparison, Before & After Adjustment

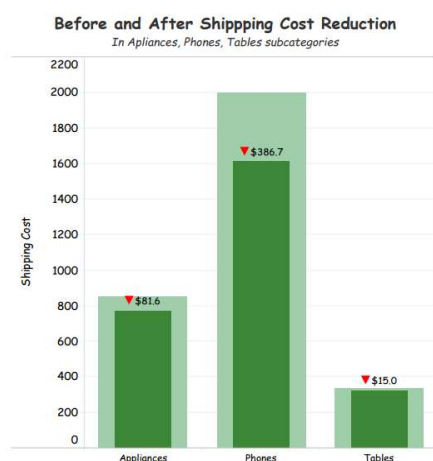


Figure 5.2 Before & After Shipping Cost Reduction

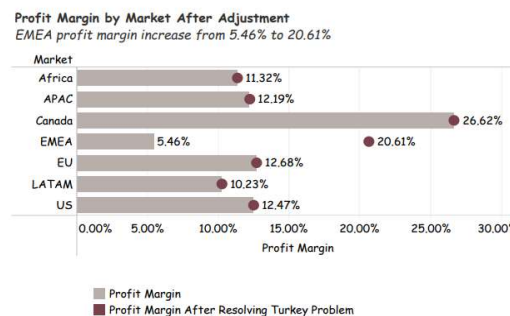


Figure 5.3 Profit Margin by Market After Adjustment

Figure 5.1 is the projected result when discount and shipping cost are adjusted. If the discount is reduced to the average of other EMEA countries which is 4.33% (figure 2.2) there will be a huge increase in profit. Especially if it's followed by lowering down 50% of shipping cost on critical and high order priority within the first class ship mode in the high shipping cost state such as Ankara, Diyarbakir, and others (figure 4.4). Then the profit will rise from -\$98,305 to \$44,202. Higher portion of the increase in profit is driven by discounts adjustment.

On figure 5.2, the shipping cost in each subcategory has dropped. Even though it's a small reduction, with a larger number of shipping costs in future, the impact of this would be significant.

Figure 5.3 shows when Turkey issues have been resolved, EMEA profit margin will increase from 5.46% to 20.61%. By following the adjustment, the objective will be met.

Recommendation

In order to improve EMEA profit margin, solving the problem in Turkey is necessary. As discounts are unable to boost sales or customer retention, it's highly recommended to reduce it. Besides that, states with higher shipping costs such as Izmir and Kocaeli are industrial areas, and Ankara is the capital city of Turkey. While Bingol and Diyarbakir are located remotely that causes higher shipping costs. In addition, higher order priority causes higher shipping cost. To overcome these issues, offering pick-up points might be cheaper and could increase customer satisfaction due to convenience. Customers can choose to pick-up from an offered location that will be beneficial when it's time-sensitive like critical or higher order priority. When inventory is not available in each state, it should be placed in a central location, to minimize the ship distances across states. Superstore could increase inventory and bulk shipping to cut down the cost without burdening customers.

Conclusion

To conclude, EMEA profit margin is the lowest across the market due to Turkey underperformance. Turkey's sales are increasing over the years, while profit is dropping. Through deeper analysis, high discount is the core issue. Shipping cost is the other factor that explains low profit margin. Tables, appliances, and phones have the highest shipping cost and ship-to-sales ratio when it's using first class mode. Critical and high order priority on first class mode incurs higher shipping cost. Shipping costs also vary by states. Lastly, there're recommendations for resolving the issue and boosting EMEA profit margin.